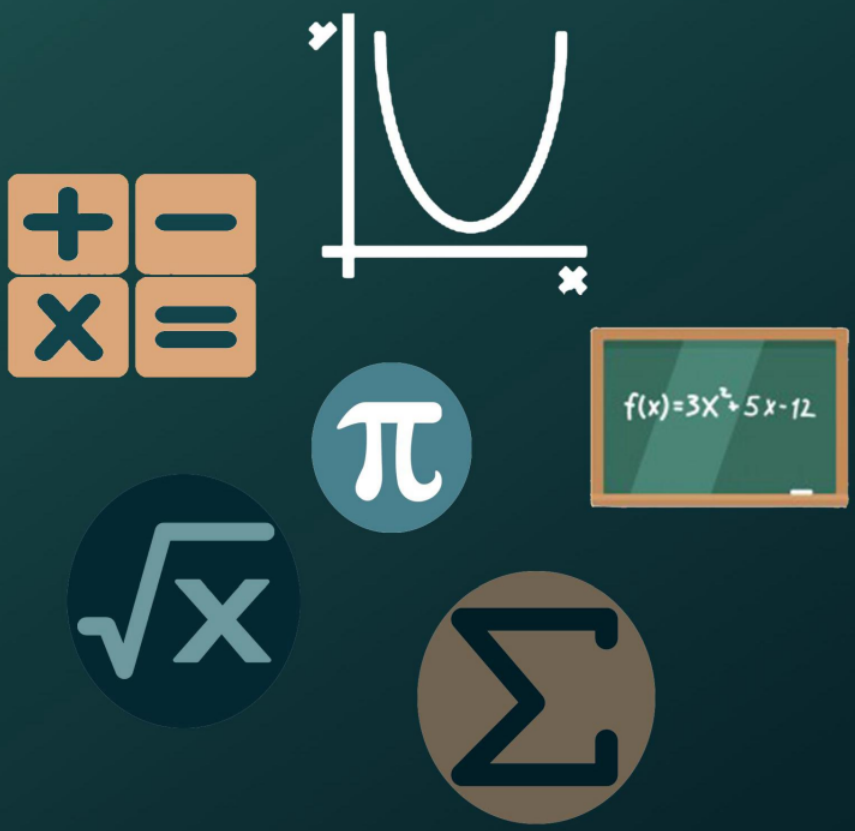
MATHEMATICS

Permutation & Combination



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Fundamental Principle of Counting	1	19-22		
2	Factorial		13		
3	Formation of Numbers	2-4	1, 14, 23	1-8	1, 2
4	Permutation and Combination	5-11	2,3,15, 24- 26	9	
5	Selection of 'r' things out of 'n' different things	12-14	4-10	10-15	3, 23, 27
6	Permutation of Things which are all Different	15,16	16, 27	16-18	24,28
7	Formation of Groups and their Distribution	17	28	19,20	8-10
8	Combination of Things which are not all Different	18-20		21	
9	Number of Divisors and their Sum	21	29,30	22	29
10	Permutation of Alike Objects Taken All or Some at a Time	22-27	17	23-25	11-13
11	Rank of the Word	28		26	25,30
12	Circular Permutation: When Objects are Different or Alike	29-32		27,28	
13	Geometric Problems	33,34	11	29-31	14-16
14	Principle of Inclusion and Exclusion				31
15	Summation of Numbers			32	26
16	Distribution of Alike Objects	35-37		33-35	20-22
17	Derangement	38,39	18		
18	Miscellaneous Problems in P&C	40,41	12	36,37	
19	After Complete Chapter				4-7,17-19

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Fundamental Principal of Counting

Q1. A number lock consist of three rings. If Ist is marked with 0 to 9 digits, 2nd ring marked with prime numbers greater than 2 but less than 30 and 3rd ring marked with all vowels, find the total number of unsuccesful attempt.

(A) 405 (B) 449
(C) 499 (D) 501

Formation of Numbers

Q2. How many numbers can be possible by using digits 3, 4, 5, 6, 7, 8 which are greater than 750000, if repetition is not allowed ?

(A) 192 (B) 120
(C) 216 (D) None

Q3. Assuming that no two consecutive digits are same, the number of n digit numbers, is :-

(A) n! (B) 9!
(C) 9^n (D) n^9

Q4. How many even numbers of three digits each can be made with the digits 1, 2, 3, 4, 6, 7 if no digit is repeated ?

(A) 75 (B) 60
(C) 36 (D) None of these

Permutation and Combination

Q5. The number of words can be formed from all the letter of the word 'MONDAY' in which first letter is a vowel are :-

(A) 120 (B) 240 (C) 360 (D) 500

Q6. Find the total number of ways of answering 5 objective type questions, each question having 4 choices.

(A) 5^4 (B) 5^4-1
(C) 4^5 (D) 4^5-1

Q7. How many different signals can be made by 5 flags from 8 flags of different colours?

(A) 7620 (B) 6720
(C) 6746 (D) None of these

Q8. In how many ways can 3 prizes be distributed among 4 boys, when no boy gets more than one prize ?

(A) 24 (B) 28 (C) 42 (D) 36

Q9. In how many ways can 3 prizes be distributed among 4 boys, when a boy may get any number of prizes ?

(A) 64 (B) 84 (C) 36 (D) 81

Q10. In a group of boys the number of arrangement of 4 boys is 12 times the number of arrangement of 2 boys. The number of boys in the group are :

(A) 10 (B) 8 (C) 6 (D) 4

Q11. If ${}^nP_5 = 20.{}^nP_3$. Then value of 'n' is equal:-

(A) 9 (B) 8
(C) 7 (D) 6

Selection of 'r' things out of 'n' different things

Q12. At an election, a voter may vote for any number of candidates, not greater than the number to be elected. There are 10 candidated and 4 are to be elected. If a voter votes for at least one candidates, then the number of ways in which he can vote is

(A) 385 (B) 1110
(C) 5040 (D) 6210

Q13. A committee consisting of 2 men and 2 women is to be chosen from 5 men and 6 women. In how many ways can this be done ?

(A) 150 (B) 25

(C) 24 (D) 360

Q14. A father with 7 children, takes 4 at a time to the zoo, as often as he can without taking the same four children together more than once. How often will he go? How often will each child go ?

(A) 30, 10 (B) 35, 15

(C) 30, 20 (D) 35, 20

Permutation of Things which are all Different

Q15. The number of ways in which the letters of the word TRIANGLE can be arranged such that two vowels do not occur together is :

(A) 1200 (B) 2400

(C) 14400 (D) None of these

Q16. Out of seven consonants and four vowels, the number of words of six letters, formed by taking four consonants and two vowels is (Assume that each ordered group of letter is a word) :

(A) 210 (B) 462

(C) 151200 (D) 332640

Formation of Groups and their Distribution

Q17. Total number of ways in which 22 different books can be given to 5 students, so that two students get 5 books each and all the remaining students get 4 books each, is equal to :-

(A) $\frac{22!}{3!2!5!(4!)^3}$ (B) $\frac{22!}{(5!)^2(4!)^3}$

(C) $\frac{22!}{(5!)^2(4!)^3(3!)}$ (D) None of these

Combination of Things which are not all Different

Q18. The number of groups that can be made from 5 different green balls, 4 different blue balls and 3 different red balls, if at least 1 green and 1 blue ball is to be included.

(A) 3700

(B) 3720

(C) 4340

(D) None of these

Q19. In how many ways atleast one horse and atleast one dog can be selected out of eight horses and seven dogs ?

(A) $2^{15} - 2$

(B) $2^{15} - 1$

(C) $(2^8 - 1)(2^7 - 1)$

(D) ${}^{15}C_2$

Q20. A shopkeeper has 10 copies of each of nine different books, then number of ways in which atleast one book can be selected is-

(A) $9^{11}-1$ (B) $10^{10}-1$

(C) 11^9-1 (D) 10^9

Number of Divisors and their Sum

Q21. The number of proper divisors of $2^p . 6^q . 15^r$ is-

(A) $(p + q + 1)(q + r + 1)(r + 1)$

(B) $(p + q + 1)(q + r + 1)(r + 1) - 1$

(C) $(p + q)(q + r)r - 2$

(D) None of these

Permutation of Alike Objects Taken All or Some at a Time

Q22. The number of words that can be made by rearranging the letters of the word "APURBA" so that vowels and consonants alternate is-

(A) 18 (B) 35

(C) 36 (D) None of these

Q23. The number of words which can be formed from letters of the word 'INSTITUTION' in which first two letters are 'N', are :-

(A) $\frac{10!}{3!3!}$ (B) $\frac{9!}{3!3!}$

(C) $\frac{11!}{3!3!2!}$ (D) $\frac{10!}{3!3!2!}$

Q24. Number of different words that can be formed from all letters of word APPLICATION such that two vowels never come together is -

(A) (45)7! (B) 8!

(C) 6!7! (D) (32)6!

Q25. Six identical coins are arranged in a row. The number of ways in which the number of tails is equal to number of heads is

(A) 20 (B) 10

(C) 9 (D) 40

Q26. The number of permutations that can be formed by arranging all the letters of the word 'NINETEEN' in which no two E's occur together is -

(A) $\frac{8!}{3!3!}$ (B) $\frac{5!}{3! \times {}^6C_2}$

(C) $\frac{5!}{3!} \times {}^6C_3$ (D) $\frac{8!}{5!} \times {}^6C_3$

Q27. There are (n + 1) white and (n + 1) black balls, each set numbered 1 to n + 1. The number of ways the balls can be arranged in a row so that adjacent balls are of different colours, is -

(A) $((n+1)!)^2$ (B) 2(2n !)

(C) 2((n + 1)!) (D) $2((n+1)!)^2$

Rank of the Word

Q28. If the letters of the word 'SACHIN' are arranged in all possible ways and these words are written out as in dictionary, then the word 'SACHIN' appears at serial number

(A) 602 (B) 603

(C) 600 (D) 601

Circular Permutation: When Objects are Different or Alike

Q29. If eleven members of a committee sit at a round table so that the. President and Secretary always sit together, then the number of arrangements is :-

(A) 10! × 2 (B) 10!

(C) 9! × 2 (D) None of these

Q30. In how many ways a garland can be made from exactly 10 flowers :-

(A) 10! (B) 9!

(C) 2(9!) (D) $\frac{9!}{2}$

Q31. There are 5 gentlemen and 4 ladies to dine at a round table. In how many ways can they seat themselves so that no two ladies are together ?

(A) 4! 5! (B) 5! 6!

(C) 4 × 5! (D) 5 × 4!

Q32. The number of ways in which 4 boys and 4 girls can stand in a circle so that each boy and each girl is one after the other is-

(A) 3! . 4! (B) 4! . 4!

(C) 8! (D) 7!

Geometric Problems

Q33. There are 12 points in a plane. The number of the straight lines joining any two of them when 3 of them are collinear is-

(A) 60 (B) 62

(C) 64 (D) 66

Q34. 'm' parallel lines in a plane are intersected by a family of 'n' parallel lines. The total number of parallelograms so formed is-

(A) $\frac{(m-1)(n-1)}{4}$

(B) $\frac{mn}{4}$

(C) $\frac{nm(m-1)(n-1)}{2}$

(D) $\frac{nm(m-1)(n-1)}{4}$

Distribution of Alike Objects

Q35. Find number of positive integral solution of equation $x + y + z = 12$ is :-

(A) ${}^{14}C_2$ (B) ${}^{11}C_2$

(C) ${}^{12}C_3$ (D) None of these

Q36. Find the number of non negative integral solution of $x + y + z + w \leq 20$ is :-

(A) ${}^{24}C_4$ (B) ${}^{20}C_4$

(C) ${}^{24}C_5$ (D) ${}^{20}C_5$

Q37. Number of ways in which 25 identical balls can be distributed among Ram, Shyam, Sunder and Ghanshyam such that atleast 1, 2, 3, and 4 balls are given to Ram, Shyam, Sunder and Ghanshyam respectively, is :-

(A) ${}^{18}C_4$ (B) ${}^{28}C_3$

(C) ${}^{24}C_3$ (D) ${}^{18}C_3$

Derangement

Q38. A person writes letters to 6 friends and addresses the corresponding envelopes. In how many ways can the letters be placed in the envelopes so that at least 4 of them are in wrong envelopes ?

(A) 72 (B) 719

(C) 664 (D) 135

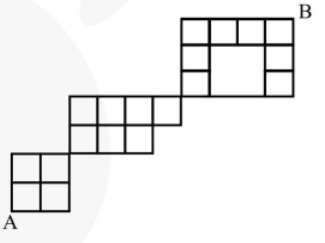
Q39. There are four balls of different colours and four boxes of same colours as of balls. Number of ways in which the balls can be placed one in each box such that the balls does not go in the box of same colour is :-

(A) 6 (B) 7

(C) 8 (D) 9

Miscellaneous Problems in P&C

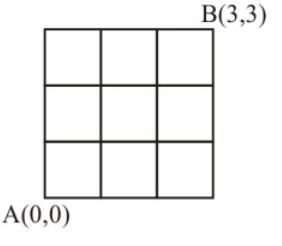
Q40. The total number of shortest path from A to B is :-



(A) 1530 (B) 1428

(C) 3150 (D) None

Q41. In how many ways can a person go from A to B if he can travel only to the right or upward along the lines?



(A) 10 (B) 20

(C) 18 (D) 16

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Consider numbers formed between 10 and 10,000 that can be formed using digits 1,2,3,4,5. If M denotes those numbers in which No digit is repeated and N denotes the number when digits can be repeated then :-

(A) M = 200 (B) M = 775

(C) N = 775 (D) N = 200

Q2. One has 10 different shirts, 5 different trousers, 5 different ties, 3 different pair of shoes. If A denotes number of different ways he can wear if wearing all items is necessary and B denotes number of different ways he can wear if shirt and trouser is compulsory then :-

(A) A = 750 (B) A = 1200

(C) B = 750 (D) B = 1200

Q3. If P(n, n) denotes the number of permuta-tions of n different things taken all at a time then P(n, n) is also identical to-

(A) r! . P(n, n – r) (B) (n – r) . P(n, r)

(C) n . P (n –1, n –1) (D) P (n, n – 1)

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions.Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques.4 to 6)

For a cricket match 11 players has to be choosen from 8 batsmen, 6 bowlers, 4 all rounders and 2 wicket-keepers to form a team. Now answer the following :

Q4. Number of selection when atmost 1 all rounder and 1 wicket keeper can play-

(A) ${}^4C_1 \cdot {}^{14}C_{10} + {}^2C_1 \cdot {}^{14}C_{10} + {}^4C_1 \cdot {}^2C_1 \cdot {}^{14}C_9 + {}^{14}C_{11}$

(B) ${}^4C_1 \cdot {}^5C_{11} + {}^{15}C_{11}$

(C) ${}^4C_1 \cdot {}^{15}C_{10} + {}^{15}C_{11}$

(D) ${}^4C_1 \cdot {}^{14}C_{10} + {}^{14}C_{10} + {}^{14}C_{11}$

Q5. No. of selection when 2 particular batsman do not want to play, if a particular bowler will play-

(A) ${}^{17}C_{10} + {}^{18}C_{11}$

(B) ${}^{17}C_{10} + {}^{19}C_{11} + {}^{17}C_{11}$

(C) ${}^{17}C_{10} + {}^{19}C_{11}$

(D) None

Q6. No. of selection when a particular batsman and a particular wicket keeper don't want to play together -

(A) $2 \cdot {}^{18}C_{10}$ (B) ${}^{19}C_{11} + {}^{18}C_{10}$

(C) ${}^{19}C_{10} + {}^{19}C_{11}$ (D) None

Paragraph # 2 (Ques. 7 to10)

There are 10 professors and 20 students in a class.

Q7. Find the number of ways in which a committee of 2 professors and 3 students is formed.

(A) 51300 (B) 53100

(C) 50310 (D) 53120

Q8. Find the number of committees in which a particular professor is included.

(A) 10260 (B) 12060

(C) 10620 (D) None of these

Q9. Find the number of committees in which a particular student is included.

(A) 7695 (B) 7965

(C) 7659 (D) 7665

Q10. Find the number of committees in

(A) 43605 (B) 43650

(C) 43600 (D) 43560

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. Four options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q11.

Column- I	Column- II
(P) Number of straight lines joining any two points out of 10 points of which four points are collinear	**(1)** 30
(Q) Maximum number of points of intersection of 10 straight lines in a plane	**(2)** 60
(R) Maximum number of point of intersection of six circles in the plane	**(3)** 40
(S) Maximum number of point of intersection of six parabolas.	**(4)** 45

	(P)	(Q)	(R)	(S)
(A)	3	4	1	2
(B)	3	1	4	2
(C)	2	1	4	3
(D)	2	4	1	3

Q12. Consider a 6 × 6 chess board then

Column- I	Column- II
(P) Number of Rectangles	**(1)** 2400
(Q) Number of squares	**(2)** 441
(R) Number of ways three square can be selected if they are NOT is same row or column	**(3)** 91
(S) Number of ways in which eleven + sign can be arranged in squares if no row remains empty.	**(4)** 252

	(P)	(Q)	(R)	(S)
(A)	2	1	3	4
(B)	2	3	1	4
(C)	1	2	3	4
(D)	2	3	4	1

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q13. Find the exponent of 6 in 50!

Q14. If λ denotes the number of 3 digit numbers with distinct digit, whose each digit is odd, then value of $\frac{\lambda}{15}$ is :-

Q15. There are 5 items in column-A and 5 items in column-B. If 'a' denotes the number of ways to match each item in column-A with an item in column-B and 'b' denotes the number of ways in which items of Column-A can be matched with one or more items of Column - B then value of $\left(\frac{a}{5!}+\frac{b}{625}\right)$ is-

Q16. The letters of the word FLOWER are taken 4 at a time and arranged in all possible ways. If the number of arrangements that contain letter F is 'a' and which do NOT contain 'F' is 'b' then $\frac{a}{b}$ is :-

Q17. Number of words that can be made by arranging letters of words 'ROORKEE' that neither begin with R nor end with E is.....

Q18. A person writes letter to six friends and address the corresponding envelopes. If λ denotes the number of ways the letter can be placed in envelopes such that all letters are in wrong envelope than $\frac{\lambda}{53}$ is:-

SUBJECTIVE TYPE

Q19. In how many ways can clean & clouded (overcast) days occur in a week assuming that an entire day is either clean or clouded.

Q20. Four visitors A, B, C & D arrive at a town which has 5 hotels. In how many ways can they disperse themselves among 5 hotels, if 4 hotels are used to accommodate them.

Q21. There are 6 roads between A & B and 4 roads between B & C.

(i) In how many ways can one drive from A to C by way of B ?

(ii) In how many ways can one drive from A to C and back to A, passing through B on both trips ?

(iii) In how many ways can one drive the circular trip described in (ii) without using the same road more than once.

Q22. (i) How many car number plates can be made if each plate contains 2 different letters of English alphabet, followed by 3 different digits.

(ii) Solve the problem, if the first digit cannot be 0.

Q23. If repetitions are not permitted

(i) How many 3 digit numbers can be formed from the six digits 2, 3, 5, 6, 7 & 9 ?

(ii) How many of these are less than 400 ?

(iii) How many are even ?

(iv) How many are odd ?

(v) How many are multiples of 5 ?

Q24. (i) In how many ways can four passengers be accommodated in three railway carriages, if each carriage can accommodate any number of passengers.

(ii) In how many ways four persons can be accommodated in 3 different chairs if each person can occupy only one chair.

Q25. Find the number of 7 lettered palindromes which can be formed using the letters from the English alphabets.

Q26. Number of ways in which 7 different colours in a rainbow can be arranged if green is always in the middle.

Q27. (i) Find the number of four letter word that can be formed from the letters of the word HISTORY. (each letter to be used atmost once)

(ii) How many of them contain only consonants?

(iii) How many of them begin & end in a consonant?

(iv) How many of them begin with a vowel?

(v) How many contain the letters Y?

(vi) How many begin with T & end in a vowel?

(vii) How many begin with T & also contain S?

(viii) How many contain both vowels?

Q28. In how many ways can 8 different books be distributed among 3 students if each receives at least 2 books ?

Q29. Find the number of proper divisors of the number 38808. Also find the sum of these divisors.

Q30. In how many ways the number 18900 can be split in two factors which are relative prime (or coprime) ?

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The number of 10-digit numbers such that the product of any two consecutive digits in the number is a prime number, is :
(A) 1024 (B) 2048
(C) 512 (D) 64

Q2. The number of different seven digit numbers that can be written using only three digits 1, 2 & 3 under the condition that the digit 2 occurs exactly twice in each number is :
(A) 672 (B) 640
(C) 512 (D) none

Q3. All possible three digits even numbers which can be formed with the condition that if 5 is one of the digit, then 7 is the next digit is :
(A) 5 (B) 325 (C) 345 (D) 365

Q4. Number of permutations of 1, 2, 3, 4, 5, 6, 7, 8 and 9 taken all at a time, such that the digit
1 appearing somewhere to the left of 2
3 appearing to the left of 4 and
5 somewhere to the left of 6, is
(*e.g.* 815723946 would be one such permutation)
(A) $9 \cdot 7!$ (B) $8!$
(C) $5! \cdot 4!$ (D) $8! \cdot 4!$

Q5. Number of odd integers between 1000 and 8000 which have none of their digits repeated, is :
(A) 1014 (B) 810
(C) 690 (D) 1736

Q6. Number of different natural numbers which are smaller than two hundred million & using only the digits 1 or 2 is :
(A) $(3).2^8 - 2$ (B) $(3).2^8 - 1$
(C) $2(2^9 - 1)$ (D) none

Q7. If m denotes the number of 5 digit numbers if each successive digits are in their descending order of magnitude and n is the corresponding figure, when the digits are in their ascending order of magnitude then $(m - n)$ has the value
(A) ${}^{10}C_4$ (B) ${}^{9}C_5$ (C) ${}^{10}C_3$ (D) ${}^{9}C_3$

Q8. Number of cyphers at the end of ${}^{2002}C_{1001}$ is :
(A) 0 (B) 1 (C) 2 (D) 200

Q9. A new flag is to be designed with six vertical strips using some or all of the colours yellow, green, blue and red. Then, the number of ways this can be done such that no two adjacent strips have the same colour is -
(A) 12×81 (B) 16×192
(C) 20×125 (D) 24×216

Q10. The number of ways in which a mixed double game can be arranged from amongst 9 married couples if no husband and wife play in the same game is–
(A) 756 (B) 1512
(C) 3024 (D) None of these

Q11. 5 Indian & 5 American couples meet at a party & shake hands . If no wife shakes hands with her own husband & no Indian wife shakes hands with a male, then the number of hand shakes that takes place in the party is :
(A) 95 (B) 110 (C) 135 (D) 150

Q12. A student has to answer 10 out of 13 questions in an examination. The number of ways in which he can answer if he must answer atleast 3 of the first five questions is :
(A) 276 (B) 267 (C) 80 (D) 1200

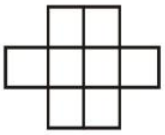
Q13. A committee of 5 is to be chosen from a group of 9 people. Number of ways in which it can be formed if two particular persons either serve together or not at all and two other particular persons refuse to serve with each other, is
(A) 41 (B) 36 (C) 47 (D) 76

Q14. A rack has 5 different pairs of shoes. The number of ways in which 4 shoes can be chosen from it, so that there will be no complete pair is :

(A) 1920 (B) 200 (C) 110 (D) 80

Q15. In the given figure of squares, 6 A's should be written in such a manner that every row contains at least one 'A'. In how many number of ways is it possible ?

(A) 24
(B) 25
(C) 26
(D) 27



Q16. In how many different ways a grandfather along with two of his grandsons and four grand daughters can be seated in a line for a photograph so that he is always in the middle and the two grandsons are never adjacent to each other.

(A) 729 (B) 600
(C) 528 (D) 848

Q17. There are six periods in each working day of a school. Number of ways in which 5 subjects can be arranged if each subject is allotted at least one period and no period remains vacant is :

(A) 210 (B) 1800
(C) 360 (D) 3600

Q18. A team of 8 students goes on an excursion, in two cars, of which one can seat 5 and the other only 4. If internal arrangement inside the car does not matter then the number of ways in which they can travel, is :

(A) 91 (B) 182 (C) 126 (D) 3920

Q19. The number of ways to give 16 different things to three persons A, B, C so that B gets 1 more than A and C gets 2 more than B is:-

(A) $\frac{16!}{4!\,5!\,7!}$ (B) $4!\,5!\,7!$

(C) $\frac{16!}{3!\,5!\,8!}$ (D) $\frac{16!\,3!}{4!\,5!\,7!}$

Q20. Number of ways in which 9 different toys be distributed among 4 children belonging to different age groups in such a way that distribution among the 3 elder children is even and the youngest one is to receive one toy more, is :

(A) $\frac{(5!)^2}{8}$ (B) $\frac{9!}{2}$

(C) $\frac{9!}{3!\,(2!)^3}$ (D) none

Q21. There are 12 balls in a basket of which 5 are red, 4 are black and 3 blue balls. Only blue balls are different, then number of ways to select atleast one ball are :-

(A) 239 (B) 119 (C) 240 (D) 120

Q22. If r, s, t are the prime numbers and p, q are the positive integers such that the LCM of p & q is $r^2t^4s^2$, then the number of ordered pair (p, q) is :

(A) 252 (B) 254 (C) 225 (D) 224

Q23. Total number of 4 letter words that can be formed from the letters of the word 'SAHARANPUR' is equal to :-

(A) 210 (B) 1237 (C) 247 (D) 1230

Q24. There are 10 red balls of different shades & 9 green balls of identical shades. Then the number of arranging them in a row so that no two green balls are together is

(A) $(10!) \cdot {}^{11}P_9$ (B) $(10!) \cdot {}^{11}C_9$
(C) $10!$ (D) $10!\,9!$

Q25. In a unique hockey series between India & Pakistan, they decide to play on till a team wins 5 matches. The number of ways in which the series can be won by India, if no match ends in a draw is :

(A) 126 (B) 252 (C) 225 (D) none

Q26. The letters of the word COCHIN are permuted and all the permutations are arranged in an alphabetical order as in an English dictionary. The number of words that appear before the word COCHIN is -

(A) 360 (B) 192 (C) 96 (D) 48

Q27. A man invites a party to (m + n) friends to dinner and places m at one particular round table and n at another. The number of ways of arranging the guest are :-

(A) $\frac{(m+n)!}{m!\,n!}$ (B) $\frac{(m+n)!}{(m-1)!(n-1)!}$

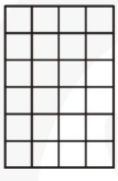
(C) (m – 1)! (n – 1)! (D) $\frac{(m+n)!}{mn}$

Q28. There are 12 guests at a dinner party. Supposing that the master and mistress of the house have fixed seats opposite one another, and that there are two specified guests who must always, be placed next to one another ; the number of ways in which the company can be placed, is:

(A) 20 . 10! (B) 22 . 10!

(C) 44 . 10! (D) none

Q29. A rectangle with sides 2m – 1 and 2n – 1 is divided into squares of unit length by drawing parallel lines as shown then number of rectangles possible with odd side lengths is -



(A) $(m + n + 1)^2$

(B) 4^{m+n-1}

(C) $m^2 n^2$

(D) mn (m + 1) (n + 1)

Q30. There are three coplanar parallel lines. If any n points are taken on each of the lines, the maximum number of triangles with vertices at these points will be

(A) $3n^2 (n-1)$ (B) $3n^2 (n-1) + 1$

(C) $n^2 (4n-3)$ (D) None of these

Q31. Lines y = x + i & y = –x + j are drawn in x – y plane such that $i \in \{1,2,3,4\}$ & $j \in \{1,2,3,4,5,6\}$. If m represents the total number of squares formed by the lines and n represents the total number of triangles formed by the given lines & x-axis, then correct option/s is/are-

(A) m + n = 50 (B) m × n = 625

(C) m + n = 48 (D) m – n = 4

Q32. Number of 7 digit numbers the sum of whose digits is 61 is :

(A) 12 (B) 24

(C) 28 (D) none

Q33. If x is an odd integer and y, z are even integers, satisfying x + y + z = 47 and x, y, $z \geq 0$, then number of solutions are -

(A) 100 (B) 200

(C) 300 (D) 400

Q34. x_1, x_2 & x_3 when divided by 4 leaves a remainder of 0,1 & 2 respectively find number of non-negative integral solution of the equation $x_1 + x_2 + x_3 = 35$, is -

(A) 45 (B) 55

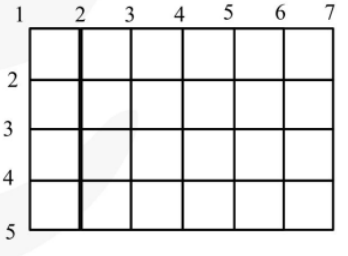
(C) 105 (D) 190

Q35. The number of ways in which 8 distinguishable apples can be distributed among 3 boys such that every boy should get atleast 1 apple & atmost 4 apples is $K \cdot {}^7P_3$ where K has the value equal to :

(A) 14 (B) 66

(C) 44 (D) 22

Q36. Number of rectangles in the grid shown which are not squares is :-



(A) 160 (B) 162

(C) 170 (D) 185

Q37. The number of ways in which two squares can be selected from an 8 by 8 chess board of size 1 × 1 so that they are not in the same row and in the same column.

(A) 2196 (B) 1024

(C) 1568 (D) 1836

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Number of 4 digit numbers of the form N = abcd which satisfy following three conditions :
(i) $4000 \le N < 6000$
(ii) N is multiple of 5
(iii) $3 \le b < c \le 6$ is equal to
(A) 12 (B) 18 (C) 24 (D) 48

Q2. Number of ways in which 3 numbers in A.P. can be selected from 1, 2, 3, ... n is :
(A) $\left(\frac{n-1}{2}\right)^2$ if n is even
(B) $\frac{n(n-2)}{4}$ if n is odd
(C) $\frac{(n-1)^2}{4}$ if n is odd
(D) $\frac{n(n-2)}{4}$ if n is even

Q3. The maximum number of permutations of 2n letters in which there are only A's & B's, taken all at a time is given by :
(A) ${}^{2n}C_n$
(B) $\frac{2}{1}\cdot\frac{6}{2}\cdot\frac{10}{3}\cdot\ldots\ldots\frac{4n-6}{n-1}\cdot\frac{4n-2}{n}$
(C) $\frac{n+1}{1}\cdot\frac{n+2}{2}\cdot\frac{n+3}{3}\cdot\frac{n+4}{4}\cdot\ldots\ldots\frac{2n-1}{n-1}\cdot\frac{2n}{n}$
(D) $\frac{2^n\cdot[1\cdot3\cdot5\ldots\ldots(2n-3)(2n-1)]}{n!}$

Q4. The combinatorial coefficient C(n, r) is equal to
(A) number of possible subsets of r members from a set of n distinct members.
(B) number of possible binary messages of length n with exactly r 1's.
(C) number of non decreasing 2-D paths from the lattice point (0, 0) to (r, n).
(D) number of ways of selecting r things out of n different things when a particular thing is always included plus the number of ways of selecting 'r' things out of n, when a particular t thing is always excluded.

Q5. There are 10 questions, each question is either True or False. Number of different sequences of incorrect answers is also equal to :
(A) Number of ways in which a normal coin tossed 10 times would fall in a definite order if both Heads and Tails are present.
(B) Number of ways in which a multiple choice question containing 10 alternatives with one or more than one correct alternatives, can be answered.
(C) Number of ways in which it is possible to draw a sum of money with 10 coins of different denominations taken some or all at a time.
(D) Number of different selections of 10 indistinguishable things taken some or all at a time.

Q6. The combinatorial coefficient ${}^{n-1}c_p$ denotes
(A) the number of ways in which *n* things of which p are alike and rest different can be arranged in a circle.
(B) the number of ways in which *p* different things can be selected out of *n* different thing if a particular thing is always excluded.
(C) number of ways in which *n* alike balls can be distributed in *p* different boxes so that no box remains empty and each box can hold any number of balls.
(D) the number of ways in which (n – 2) white balls and p black balls can be arranged in a line if black balls are separated, balls are all alike except for the colour.

Q7. Number of ways in which the letters of the word 'B U L B U L' can be arranged in a line in a definite order is also equal to the

(A) Number of ways in which 2 alike Apples and 4 alike Mangoes can be distributed in 3 children so that each child receives any number of fruits.

(B) Number of ways in which 6 different books can be tied up into 3 bundles, if each bundle is to have equal number of books.

(C) Coefficient of $x^2y^2z^2$ in the expansion of $(x + y + z)^6$.

(D) Number of ways in which 6 different prizes can be distributed equally in three children.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 8 to 10)

16 players P_1, P_2, P_3,.....P_{16} take part in a tennis tournament. Lower suffix player is better than any higher suffix player. These players are to be divided into 4 groups each comprising of 4 players and the best from each group is selected for semifinals.

Q8. Number of ways in which 16 players can be divided into four equal groups, is

(A) $\frac{35}{27}\prod_{r=1}^{8}(2r-1)$ (B) $\frac{35}{24}\prod_{r=1}^{8}(2r-1)$

(C) $\frac{35}{52}\prod_{r=1}^{8}(2r-1)$ (D) $\frac{35}{6}\prod_{r=1}^{8}(2r-1)$

Q9. Number of ways in which they can be divided into 4 equal groups if the players P_1, P_2, P_3 and P_4 are in different groups, is :

(A) $\frac{(11)!}{36}$ (B) $\frac{(11)!}{72}$

(C) $\frac{(11)!}{108}$ (D) $\frac{(11)!}{216}$

Q10. Number of ways in which these 16 players can be divided into four equal groups, such that when the best player is selected from each group, P_6 is one among them, is (k) $\frac{12!}{(4!)^3}$.

The value of k is-

(A) 36 (B) 24

(C) 18 (D) 20

Paragraph # 2 (Ques. 11 to 13)

Consider the word W = MISSISSIPPI

Q11. If N denotes the number of different selections of 5 letters from the word W = MISSISSIPPI then N belongs to the set

(A) {15, 16, 17, 18, 19}

(B) {20, 21, 22, 23, 24}

(C) {25, 26, 27, 28, 29}

(D) {30, 31, 32, 33, 34}

Q12. Number of ways in which the letters of the word W can be arranged if atleast one vowel is separated from rest of the vowels

(A) $\frac{8!\cdot 161}{4!\cdot 4!\cdot 2!}$ (B) $\frac{8!\cdot 161}{4\cdot 4!\cdot 2!}$

(C) $\frac{8!\cdot 161}{4!\cdot 2!}$ (D) $\frac{8!}{4!\cdot 2!}\cdot\frac{165}{4!}$

Q13. If the number of arrangements of the letters of the word W if all the S's and P's are separated is (K)$\left(\frac{10!}{4!\cdot 4!}\right)$, then K equals -

(A) $\frac{6}{5}$ (B) 1

(C) $\frac{4}{3}$ (D) $\frac{3}{2}$

Paragraph # 3 (Ques. 14 to 16)

There is a convex polygon of n sides (n > 8). Number of quadrilateral having three, two and one side common with polygon are represented by A, B and C respectively, Also D represents the number of quadrilateral having vertices at the vertices of polygon and at least one side common with sides of polygon, then answer the following questions:

Q14. The value of B is –

(A) (n – 5)n (B) n(n – 3)

(C) n(n – 4) (D) n(n – 6)

Q15. The value of C is given by -

(A) $\frac{n(n-3)(n-4)}{2}$

(B) $\frac{n(n-4)(n-5)}{2}$

(C) $\frac{n(n-5)(n-6)}{2}$

(D) None of these

Q16. If n = 10 then the value of C + D is-

(A) 240 (B) 360

(C) 420 (D) 540

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17.

Column-I	**Column-II**
(A) Number of increasing permutations of *m* symbols are there from the *n* set numbers $\{a_1, a_2, \ldots, a_n\}$ where the order among the numbers is given by $a_1 < a_2 < a_3 < \ldots a_{n-1} < a_n$ is	(P) n^m
(B) There are *m* men and *n* monkeys. Number of ways in which every monkey has a master, if a man can have any number of monkeys	(Q) ${}^mC_n$
(C) Number of ways in which *n* red balls and $(m-1)$ green balls can be arranged in a line, so that no two red balls are together, is (balls of the same colour are alike)	(R) ${}^nC_m$
(D) Number of ways in which '*m*' different toys can be distributed in '*n*' children if every child may receive any number of toys, is	(S) m^n

Q18.

Column-I	Column-II
(A) Four different movies are running in a town. Ten students go to watch these four movies. The number of ways in which every movie is watched by atleast one student, is (Assume each way differs only by number of students watching a movie)	(P) 11
(B) Consider 8 vertices of a regular octagon and its centre. If T denotes the number of triangles and S denotes the number of straight lines that can be formed with these 9 points then the value of (T – S) equals	(Q) 36
(C) In an examination, 5 children were found to have their mobiles in their pocket. The Invigilator fired them and took their mobiles in his possession. Towards the end of the test, Invigilator randomly returned their mobiles. The number of ways in which at most two children did not get their own mobiles is	(R) 52
(D) The product of the digits of 3214 is 24. The number of 4 digit natural numbers such that the product of their digits is 12, is	(S) 60
(E) The number of ways in which a mixed double tennis game can be arranged from amongst 5 married couple if no husband & wife plays in the same game, is	(T) 84

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q19.

Column- I	**Column- II**
(P) Total Number of selections of fruits atleast one which can be made from 3 Bannanas 4 apples and 2 oranges. (fruits of same species are alike)	**(1)** 70
(Q) If 7 points out of 12 points are collinear, then number of triangles formed are	**(2)** 59
(R) The number of ways of selecting 10 Balls from unlimited red, black white and green balls is (Balls of same colour are alike)	**(3)** 1857
(S) The total number of proper positive integers divisors of 38808 is	**(4)** 286

	(P)	**(Q)**	**(R)**	**(S)**
(A)	2	4	1	3
(B)	1	2	3	4
(C)	2	3	4	1
(D)	1	3	4	2

TRIPLE MATCH COLUMN TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Consider six objects to be distributed among four persons.

Column - 1 : Contains conditions for distribution of objects.

Column - 2 : Contains the numbe of ways of distribution when objects are distinct.

Column - 3 : Contains the number of ways of distribution when objects are identical.

Column - 1	Column - 2	Column - 3
(I) Any one gets any number of objects	(i) 4092	(P) 84
(II) Each gets atleast one object	(ii) 480	(Q) 10
(III) Each gets odd number of objects	(iii) 1560	(R) 4
(IV) No one gets all the objects	(iv) 4096	(S) 80

Q20. Which of the following is NOT a correct combination :-

(A) (I) (iv) P (B) (III) (iii) Q (C) (III) (ii) R (D) (IV) (ii) S

Q21. Which of the following is NOT a CORRECT combination :-

(A) (IV) (i) (S) (B) (III) (iii) (Q) (C) (I) (iv) (P) (D) (II) (iii) (Q)

Q22. Which of the following is NOT a correct combination :-

(A) (I) (iv) (P) (B) (I) (ii) (Q) (C) (III) (ii) (R) (D) (IV) (i) (S)

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q23. In a certain algebraical exercise book there are 4 examples on arithmetical progressions, 5 examples on permutation-combination and 6 examples on binomial theorem . Number of ways a teacher can select for his pupils atleast one but not more than 2 examples from each of these sets, is _____.

Q24. If number of ways in which elements of set $S = \{1, 2, 3..... 12\}$ is partitioned into three sets A, B, C of equal size such that $A \cap B = B \cap C = C \cap A = \phi$ is $\frac{a!}{(b!)^3}$ then (a – b) is-

Q25. A cricket team consists of 10 pairs of twin brothers. On the first day of training, all 20 players stand in a circle in such a way that all pairs of twin brothers are neighbours. Numbers of ways this can be done is $(k!)2^{k+1}$ then 'k' equals =

Q26. If k denotes the sum of all natural numbers 'n' such that $100 \le n \le 4000$ that can be made with the digits 0,1,2,3,4. If repetition of digits in same number is allowed then number of cyphers in k is :-

SUBJECTIVE TYPE

Q27. Six married couple are sitting in a room. Find the number of ways in which 4 people can be selected so that

(i) they do not form a couple

(ii) they form exactly one couple

(iii) they form at least one couple

(iv) they form atmost one couple

Q28. Find the number of ways in which the letters of the word "MIRACLE" can be arranged if vowels always occupy the odd places.

Q29. Find the total number of proper factors of the number 35700. Also find

(i) sum of all these factors,

(ii) sum of the odd proper divisors,

(iii) the number of proper divisors divisible by 10 and the sum of these divisors.

Q30. There are 720 permutations of the digits 1, 2, 3, 4, 5, 6. Suppose these permutations are arranged from smallest to largest numerical values, beginning from 1 2 3 4 5 6 and ending with 6 5 4 3 2 1.

(i) What number falls on the 124th position?

(ii) What is the position of the number 321546?

Q31. Find the number of permutations of letters a,b,c,d,e,f,g taken all at a time if neither 'beg' nor 'cad' pattern appear.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
B	A	C	B	B	C	B	A	A	C
11	12	13	14	15	16	17	18	19	20
B	A	A	D	C	C	A	B	C	C
21	22	23	24	25	26	27	28	29	30
B	C	B	A	A	C	D	D	C	D
31	32	33	34	35	36	37	38	39	40
A	A	C	D	B	A	D	C	D	B
41									
B									

Ex - 1A

1. A,C **2.** A,D **3.** A,C,D **4.** A **5.** C

6. B **7.** A **8.** A **9.** A **10.** A

11. A **12.** B **13.** 22 **14.** 4 **15.** 6

16. 2 **17.** 330 **18.** 5 **19.** 128 **20.** 120

21. (i) 24 (ii) 576 (iii) 360 **22.** (i) 468000 (ii) 421200

23. (i) 120 (ii) 40 (iii) 40 (iv) 80 (v) 20

24. (i) 3^4 (ii) 24 **25.** 26^4 **26.** 720

27. (i) 840 (ii) 120 (iii) 400 (iv) 240 (v) 480 (vi) 40 (vii) 60 (viii) 240

28. $\left[\frac{8!}{(2!)^2 4!2!}+\frac{8!}{(3!)^2 2!2!}\right]\times 3!$ **29.** (i) 71 (ii) 94572 **30.** 8

Ex-2

1	2	3	4	5	6	7	8	9	10
B	A	D	A	D	A	B	B	A	B
11	12	13	14	15	16	17	18	19	20
C	A	A	D	C	C	B	C	A	C
21	22	23	24	25	26	27	28	29	30
A	C	D	B	A	C	D	A	C	C
31	32	33	34	35	36	37			
A	C	C	A	D	A	C			

Ex - 2A

1. C **2.** C,D **3.** A,B,C,D **4.** A,B,D **5.** B,C,D

6. B,D **7.** A,C,D **8.** A **9.** C **10.** D

11. C **12.** B **13.** B **14.** A **15.** B

16. B

17. (A)→(R); (B)→(S); (C)→(Q); (D)→(P)

18. (A)→(T); (B)→(R); (C)→(P); (D)→(Q); (E)→(S)

19. C **20.** D **21.** B **22.** B **23.** 3150

24. 8 **25.** 9 **26.** 1

27. (i) 240 (ii) 240 (iii) 255 (iv) 480 **28.** 576

29. (i) 89291 (ii) 17855 (iii) 67980 **30.** (i) 213564 (ii) 267th

31. 7! – 5! – 5! + 3!

-: Important Notes & Formulas :-